MINI ASSESSMENT

E-COMMERCE API RELIABILITY & SAFE DATA HANDLING

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1 **An order API returns status 503. How should the application respond?**

PROCESS FLOW

Order Request	**503 Service Unavailable**	**Short Backoff & Retry**	**Limit Retries Safely**

Answer	A 503 response usually means the service is temporarily unavailable. The application should wait briefly and retry using a controlled backoff strategy. It should cap the number of retries and return a clear failure message if the service remains unavailable.

2 **What should a program do when an API response is not valid JSON?**

PROCESS FLOW

Response Received	**Try JSON Parsing**	**Catch Parse Error**	**Fallback / Report Safely**

Answer	The program should catch the JSON parsing exception instead of crashing. It can inspect the HTTP status and response text for diagnosis, then either use a safe fallback or stop the current operation in a controlled way.

SAFE ACCESS TO OPTIONAL ORDER DATA

3 **Why is get() safer than direct key access when a JSON field may be absent?**

PROCESS FLOW

JSON Field Optional	**Use get("field")**	**Return Value or Default**	**Avoid Key Error**

Answer	Using data.get("field") is safer when the field is optional because it returns None or a supplied default when the key is missing. Direct access such as data["field"] raises a KeyError if the field does not exist.

KEY TAKEAWAY

Reliable API clients should expect temporary service failures, malformed responses, and optional fields. Controlled retries, exception handling, and safe JSON access help keep applications stable.